Kyle Sabado

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Education

University of Toronto — Sept. 2022 – May 2027
Bachelor of Applied Science in Computer Engineering, Minor in Robotics — *Toronto, ON*
Relevant Courses: Matrix Algebra & Optimization, Probabilistic Reasoning, Real Analysis, Data Structures & Algorithms, Operating Systems, Computer Networks, Machine Learning, Applied Deep Learning

Technical Skills

Languages: Python, C++, C, MATLAB, Java, Verilog, Bash Scripting, HTML/CSS
Tools & Services: Git, Linux, Docker, Robot Operating System (ROS), Google Cloud, Visual Studio
Libraries & Software: PyTorch, vLLM, Ray, NCCL, Weights & Biases, HuggingFace, OpenCV

Work Experience

AI Networking Intern — May 2025 – April 2026
Huawei Canada — *Toronto, ON*
- Optimized inference performance of **Mixture-of-Experts** (MoE) LLMs by **20%** through a custom topology-aware system in Python, implementing expert sharding and replication to minimize inter-node communication overhead.
- Developed a scalable **collective communication scheduler** for multi-tenant AI workloads in GPU clusters using **Python** and TCP/RPC, improving training throughput by **30%** in high contention scenarios.
- Implemented an expert selection algorithm for **Expert Parallel Load Balancing (EPLB)** in **vLLM**, minimizing network latency due to token routing.

Research Experience

Reinforcement Learning Research Intern — May 2026 – Present
MARMoT Lab, National University of Singapore — *Singapore*
- Researching continuous-space multi-agent pathfinding by combining expert planning with residual reinforcement learning.
- Developed a multi-agent controller using **PPO** to learn residual corrections over **LaCAM3** expert trajectories, achieving up to a **99%** success rate and an **8%** reduction in makespan.

Wireless Networks Research Student — May 2024 – Sept. 2024
WIRLab, University of Toronto — *Toronto, ON*
- Researched **RIS**-enabled localization in wireless networks, achieving **centimeter-level** tracking accuracy; co-authored a paper accepted at **IEEE WCNC 2025**.
- Designed and simulated **EKF**-based tracking algorithms in **MATLAB** for precise localization of user equipment (UEs), reducing position estimation error in multipath environments.

Projects

Deep Learning Skin Cancer Classifier | *Python, PyTorch, Scikit-learn* — May 2025 – Aug. 2025
- Built an **ensemble CNN model** in a team of 4, outperforming individual model baselines by **+5% accuracy** on skin lesion image classification.
- Applied **data augmentation**, **transfer learning** and **class rebalancing** to mitigate dataset imbalance, improving generalization and achieving **80% balanced accuracy** on the HAM10000 dataset.

Mapping Application | *C++* — Sept. 2023 - May 2024
- Collaborated in a team of 3 to build a full-stack mapping application in **C++** using the **OpenStreetMap API**, enabling interactive navigation and route visualization.
- Implemented the **A* algorithm** for efficient single-destination pathfinding and a custom **greedy heuristic** for traveling salesman-like optimization, reducing route computation time.

Publications

M. Ammous*, **K. Sabado***, M. Saif and S. Valaee, "Cooperative Localization and Tracking Using RISs and Sidelink Communications," 2025 IEEE Wireless Communications and Networking Conference (WCNC), Milan, Italy, 2025.